A Hypothesis Engine over History: Combinatorial Generation, Subjective-Logic Belief, and Structural Unknowns*

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Abstract

History accumulates claims faster than any framework scores them: a correlation, a dated fragment, a contested attribution each carry their own confidence number, with no credence a reader can compare across hypotheses. We present a design for a hypothesis engine that treats every placement of an actor, action, and mechanism into a dated interval, and every ordered pair of placements, as one point in a combinatorial address space, generated whether or not the literature supports it. Each hypothesis carries a subjective-logic opinion, belief, disbelief, and uncertainty mass, rather than one point estimate, so a hypothesis nobody has examined reads apart from one refuted. A structural-unknowns layer puts the engine's blind spots, an unnamed concept, an unexcavated gap, undiscovered evidence, inside the model instead of a footnote added later. We derive the address scheme's size and injectivity, the opinion model's normalization and probability bound, and lay out an engine loop of generation, critique, ranking, and evolution, checked against a sibling engine for a different corpus. We close with a calibration design, two holdout tests and two candidate pilot periods, naming what in the model stays uncalibrated. No results are claimed here; this is a design and the proofs that check it.

1 Introduction

A corpus of historical claims grows by accretion: a correlation between two flood myths, a radiocarbon date on a construction fill, a contested reading of an inscription. Four disjoint scores can sit on such a claim, an embedding similarity, a source-quality heuristic, a tier-classifier softmax, a keyword count, and none answers the question a reader asks: how likely is this account of the past to be true. Closing that gap raises three problems at once, and a design that solves only one of them does not close it.

First, the set of hypotheses worth scoring cannot be bounded by what published sources already argue. A corpus with a coherent radiocarbon date and a coherent construction technique supports the reading that farmers built a shrine at a Neolithic site; the same corpus, read against the same address space, also supports the reading that an unnamed actor with an unnamed mechanism did it, at a prior so low no scholar would publish it and no engine should refuse to name it. An engine that only proposes what a literature review would propose inherits that literature's blind spots along with its evidence.

Second, a posterior collapsed to one number cannot tell a reader whether a hypothesis sits at even odds because two strong, opposed evidence clusters cancel, or because nobody has looked.

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A sigmoid score $\tau \in [0, 1]$ answers both cases with the same number near 0.5. A reader deciding whether to go dig, translate, or read needs the model to state which case it is in.

Third, a generator that only reports on the hypotheses it materializes has no account of what it has not materialized. A closed vocabulary of actors, mechanisms, and places has no address for a concept nobody has named yet; a hypothesis space with 10^{24} points has no process that visits all of it; a claim of "the full set" over a space that large is a claim no process on file can back.

This paper answers each problem with one piece of machinery, over Sections 4 through 9, and derives the part of each piece that can be derived rather than measured. The five contributions are:

1. A combinatorial address scheme, a fixed-prime factorization over a hypothesis's slot vocabulary, that names every placement and sequence hypothesis a corpus's vocabulary can express, decodable by trial division alone, without materializing the space it addresses (5).

2. A subjective-logic opinion model that carries belief, disbelief, and uncertainty mass apart from one another, proved to sum to one and to project to a probability in $[0, 1]$, reducing to the corpus's existing sigmoid posterior at the limit of total ignorance (6).

3. A structural-unknowns layer, gap nodes ranked by value of information, open-world slots, missing-mass estimation, and prior-profile robustness, that gives the engine's own blind spots a place in the model rather than a footnote a human adds later (7).

4. An engine loop, generation, critique, ranking tournament, evolution, and self-report, adapted from the AI co-scientist pattern [1] and cross-walked line by line against `scientific-discovery`, a sibling engine built for a different corpus and cited throughout as a software repository rather than a published article (8).

5. A calibration design, two holdout splits and two candidate pilot periods stated as design options with no result claimed, naming the constants this engine still runs on uncalibrated instead of leaving them hidden (9).

2 Related work

This section places each of the five contributions of 1 against the closest published system it extends, departs from, or stands in for.

AI hypothesis generation. A generate, critique, rank, evolve loop over a claims graph, with Elo seeded from a Bayesian posterior and updated by simulated debate, is the shape DeepMind's AI co-scientist runs over biomedical literature [1]; 8 follows the same shape over historical correlations. AlphaEvolve mutates a program population against a programmatic fitness function [2], generalizing FunSearch's evolution of one function toward a numeric objective [3]; neither has a crisp fitness function for a qualitative historical claim, so 6's belief model stands in its place. Sakana's AI Scientist runs an idea-to-code-to-experiment-to-paper-draft loop scored by an automated reviewer [4]; its v2 successor replaces that loop with an agentic tree search and submitted three fully autonomous manuscripts to a peer-reviewed ICLR workshop, one of which cleared the human acceptance threshold [5]. Stanford's Virtual Lab specializes agent roles under a principal-investigator agent and budgets a small human-review pass over its output [6], the pattern 9's founder-review budget reuses. MIT's SciAgents walks a roughly thousand-paper graph with ontologist, scientist, and critic roles [7], the role-specialization pattern 8 extends with an unknown-unknown generator and a preservation critic. FutureHouse's Robin closes a literature-to-hypothesis-to-experiment loop against a real wet-lab result, validating one candidate, ripasudil, for dry age-related macular degeneration and generating hypotheses across ten additional diseases [8]; with no wet lab to close against, 9's holdout splits stand in. MC-NEST extends Monte Carlo tree search with an explicit explore or exploit control and an LLM judge scoring novelty, clarity, significance, and verifiability [9], the control 8's ranking tournament borrows.

Belief representation. Dempster and Shafer's theory of evidence assigns belief mass to subsets of a hypothesis space rather than to single outcomes [10, 11]; Jøsang's subjective logic specializes that theory to a single binomial opinion (b, d, u, a) with an explicit uncertainty mass and a closed-form cumulative fusion rule [12], the exact form 6 builds the engine's posterior on top of.

Chronology. Bronk Ramsey's Bayesian phase modeling over stratigraphy and the radiocarbon calibration curve, the method behind the OxCal program, turns a set of dated observations into a posterior density over a period's true date [13]; Allen's thirteen qualitative interval relations give the vocabulary for ordering two such dated periods without committing to exact dates [14]. 4 builds the timeline substrate on both.

Computational history. Ithaca restores, dates, and places damaged ancient inscriptions as ranked, evidence-grounded hypotheses, trained on 176,861 inscriptions and validated against 23 historians [15]; Aeneas extends the same approach to contextualizing and attributing longer passages of ancient text [16]. Both ground a hypothesis in retrieved primary evidence with a stated confidence, the pattern 6's evidence items carry forward. Seshat codes complex societies into a per-polity, per-century variable databank built for testing theories of social evolution [17]; a recent benchmark on that databank, referred to throughout the intake specification this paper implements as Hist-LLM, reports current language models near 46 percent recall on Seshat's coded judgments (`IDEAL-STATE-AND-UNKNOWNS-SPEC.md` §7). That benchmark has no resolvable DOI or arXiv identifier, so it is not a bibliography entry here; 8 and 10 cite its finding by pointing at the specification section that reports it, the rule this paper's glossary also follows for a term with no external citable source.

Species-richness estimators. Good's population-frequency estimator gives the unseen share of a sampled population as f_1/N, the count of once-seen classes over the total sample [18]; Colwell and Coddington's extrapolation review restates that estimator's richness form, commonly called Chao1, as $S_{\text{est}} = S_{\text{obs}} + f_1^2/(2f_2)$ [19], building on Chao's own capture-recapture estimator for sparse observation data [20]. 7 applies both to a generator's own output instead of a sampled ecological population.

Dirichlet process. Ferguson's nonparametric prior over an unbounded set of categories gives a closed-form probability that the next draw names a category not yet seen, $\alpha/(\alpha + N)$ [21]; 7's open-world slot reserves exactly this mass for a concept the ontology has not yet added.

Gödel numbering. Gödel's proof that a formal system's own statements can be arithmetized as numbers, decodable by the fixed factorization of a fixed base sequence [22], is the same move 5's address scheme makes over a hypothesis's slot values in place of a formula's symbols.

Sibling engine. `scientific-discovery` (https://github.com/gianyrox/quantum-algorithm-discovery, v0.11) is a software repository with no DOI, arXiv identifier, or Zenodo archive attached; per `papers/PAPER-STANDARDS.md`'s citation rule it is cited here by URL rather than entered into the bibliography. It retrieves papers, extracts a derived research object from each, searches for structural families across disciplines, and maps surviving families against known quantum algorithms. The history engine runs the same four-stage shape, retrieve, extract, search, target, over a claims graph instead of a paper corpus; `CROSSWALK-QUANTUM-ALGORITHM- DISCOVERY.md` works the correspondence module by module, and Sections 7 and 8 carry forward eight patterns that crosswalk pulls into this design.

3 Preliminaries

This section defines every term 4 through 9 depends on, each linked via `\term` to a glossary entry (C) and a Lean counterpart (A). 1 collects the symbols carried across every section below as a single reference, so a symbol's meaning does not have to be traced back through whichever section introduced it.

Table 1: Symbols carried across 4 through 9, with the equation or definition that first fixes each.

Symbol	Meaning
a, b, d, u	Base rate, belief, disbelief, uncertainty mass of one opinion $\omega(h)$ (3.8)
P	Projected probability $P(h) = b(h) + a(h)\,u(h)$ (3.9)
r, s	Pooled supporting and refuting evidence weight for h (6.1)
W	Total-ignorance weight, fixed at 2 (11)
$D(n)$	Diminishing-returns discount on n corroborating clusters, $\lambda = 0.5$ (9)
$X(K)$	Cross-kind independence bonus over K evidence kinds (10)
μ	The corpus's original contradiction-penalty weight, 0.5; superseded by 11
δ	Detectability, $\delta(\text{period}, \text{medium}, \text{region}) \in [0, 1]$ (3.11)
τ	The corpus's pre-opinion sigmoid posterior, $\tau = \text{sigmoid}(L)$ (1)
$L(h)$	Ranking score, the logit of $P(h)$ shifted by $\Theta_{\text{temporal}}(h)$ (12)

Definition 3.1 (Astronomical-year interval)**.** The engine's time axis is a signed integer line in astronomical years: year 0 exists, and there is no gap between 1 BCE and 1 CE the way the historical BC/AD count has one. An interval $I = (\text{start}, \text{end}, \text{uncertainty})$ pairs a start and end year with a distribution over each, point, uniform, normal, or an OxCal-style posterior carrying its own sampled density curve [13]. The astronomical year axis sits under every dated node the engine holds. Lean counterpart: `Bucket.Timeline.Interval`.

Definition 3.2 (Allen relation)**.** An Allen relation is one of the thirteen qualitative relations between two intervals I_1 and I_2: BEFORE, AFTER, MEETS, MET-BY, OVERLAPS, OVERLAPPED-BY, STARTS, STARTED-BY, FINISHES, FINISHED-BY, DURING, CONTAINS, EQUALS [14]. An edge between two dated nodes carries exactly one such relation plus a confidence and a provenance pointer to the date claim it derives from. Lean counterpart: `Bucket.Timeline.AllenRelation`.

Definition 3.3 (Concept and slot)**.** A *slot* is one of `ACTOR`, `ACTION`, `OBJECT`, `PLACE`, `TIME_BIN`, `MECHANISM`, or `RELATION`. A concept c is a member of one slot's vocabulary, carrying a prior logit $\ell(c) \in \mathbb{R}$, a consensus status on the scale majority-scholarly through non-consensus, and a rationale string. Every slot's vocabulary carries an `OTHER` concept with reserved probability mass, per 7. Lean counterpart: `Bucket.Concept.Slot`.

Definition 3.4 (Placement hypothesis)**.** A placement hypothesis is one point in the product space

$$H = \texttt{ACTOR} \times \texttt{ACTION} \times \texttt{OBJECT} \times \texttt{PLACE} \times \texttt{TIME\_BIN} \times \texttt{MECHANISM}, \tag{1}$$

read as the claim that the named actor performed the named action on the named object at the named place, inside the named time bin, by the named mechanism. Lean counterpart: `Bucket.Hypothesis.Placeme` which carries the full `Interval` the corpus stores for a placement rather than the bucketed `TIME_BIN` identifier this product space addresses; `Bucket.Address.SlotTuple` (5) is the Lean structure that names `timeBin` as its own field, the one 3 encodes.

Definition 3.5 (Sequence hypothesis)**.** A sequence hypothesis is one point in

$$H_{\text{seq}} = H \times \texttt{ALLEN\_RELATION} \times H, \tag{2}$$

read as two placements joined by one of the thirteen Allen relations of 3.2. Every causal or co-occurrence hypothesis is a sequence hypothesis read through a fixed mapping from Allen relation to causal reading. Lean counterpart: `Bucket.Hypothesis.Sequence`.

Definition 3.6 (Address)**.** Fix the first thirteen primes $p_1, \ldots, p_{13} = 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41$ in slot order, six for a placement, one for the relation, six more for a sequence's second placement. Index each slot's vocabulary from 0. The address of a slot assignment is

$$\text{address}(\text{slots}) = \prod_k p_k^{v_k(\text{slots})+1}, \tag{3}$$

where $v_k(\text{slots})$ is the vocabulary index of the value assigned to the k-th slot in the fixed slot order. Lean counterpart: `Bucket.Address.encode`.

Definition 3.7 (Evidence item)**.** An evidence item is a tuple $(\kappa, \theta, \sigma, \pi, e)$: a kind κ drawn from the nine modalities of 6, a tier $\theta \in \{\text{T1}, \ldots, \text{T6}\}$ carrying a fixed weight $k(\theta)$, a span σ naming the character range and hypothesis field it grounds, a provenance record π naming who or what asserted it, and an edge strength $e \in [0, 1]$. An evidence item has no independent Lean type of its own in this pass: its definitional weight, kind, tier, and strength, is exactly what 8's $s_i = k(\theta_i) \cdot e_i$ reduces it to before it reaches `Bucket.Belief.fromEvidence`'s r and s arguments.

Definition 3.8 (Opinion)**.** An opinion is a tuple $\omega(h) = (b(h), d(h), u(h), a(h))$: belief, disbelief, uncertainty mass, and base rate, over a hypothesis h. Lean counterpart: `Bucket.Belief.Opinion`.

Definition 3.9 (Projected probability)**.** The projected probability of an opinion $\omega(h) = (b, d, u, a)$ is

$$P(h) = b(h) + a(h) \cdot u(h). \tag{4}$$

Lean counterpart: `Bucket.Belief.project`.

Definition 3.10 (Stemma and effective count)**.** A stemma over a set of evidence sources is a directed graph with one node per source and an edge `copies_from` or `shares_archetype_with` carrying a copy-confidence weight in $[0, 1]$. Fix a threshold θ_{prune} (default 0.6); the effective count n_{eff} of a cluster is the number of connected components of the stemma after removing every edge below θ_{prune}. n_{eff} replaces the raw cluster count n everywhere a diminishing-returns discount is applied. No Lean counterpart is named for this pass: connected-component counting over a pruned stemma graph is not yet formalized, and n_{eff} reaches `Bucket.Belief.fromEvidence` only as a plain `Nat` substituted for n in 9 before that function is called. 10 states this as an open item in the Lean listings alongside detectability and gap nodes.

Definition 3.11 (Detectability)**.** Detectability $\delta(\text{period}, \text{medium}, \text{region}) \in [0, 1]$ is the probability that evidence of a true hypothesis would have survived and been found in that period, medium, and region. No Lean counterpart is named for this pass; 10 states this as an open item in the Lean listings.

Definition 3.12 (Gap node)**.** A gap node is a first-class node for something the corpus has not yet looked at, an unexcavated site, an untranslated text, an undated stratum, an unread archive, carrying the set of hypotheses it would move and a value-of-information score. No Lean counterpart is named for this pass; 10 states this as an open item in the Lean listings.

Definition 3.13 (Missing mass)**.** The missing mass of a repeated generation process is the Good-Turing estimate f_1/N, where f_1 is the count of hypotheses observed in exactly one run and N is the total observed across runs [18]. The matching richness estimate is the Chao1 estimator $S_{\text{est}} = S_{\text{obs}} + f_1^2/(2f_2)$, with f_2 the count observed in exactly two runs [19, 20]. Lean counterpart: `Bucket.Unknowns.missingMass`.

4 Timeline model

Every hypothesis in this engine sits on the axis of 3.1. A historical year n BCE maps to astronomical year $-(n-1)$; n CE maps to astronomical year n. Calendar systems, Julian, Gregorian, Hebrew, Hijri, and the rest, map onto this axis through a converter per calendar, and a contested conversion becomes a dating claim surfaced alongside the event it dates rather than resolved silently in code.

A resolution ladder buckets any point on the axis at five widths: year, decade, century, millennium, era, so a hypothesis states the precision it carries instead of a false year-level exactness. An event or period node's date carries the uncertainty distribution of 3.1 at whichever width its dating evidence supports; an OxCal-posterior distribution carries the full sampled density curve behind the highest-density interval a period node reports.

Nodes nest four levels deep, era, period, event, year, each level holding a `parent` pointer to the level above. A period can carry more than one `parent` when two chronologies frame it differently, a regional and a global periodization claiming overlapping centuries, so a period node sits under two eras at once rather than forcing a single framing to win by default.

An edge between two dated nodes is a claim like any other: it carries one of the thirteen 3.2 relations, a confidence, and a `derived_from` pointer to the date posterior that supports it. 1 shows the resulting per-bin timeline export: each time bin holds its ranked placement hypotheses as bars, and an Allen relation drawn between two bins is a sequence hypothesis read directly off the same axis. The Younger Dryas and Early Holocene bins in the figure carry the exact intervals and posteriors this paper's worked examples use in Sections 6 and 9; the relation drawn between them, BEFORE at confidence 0.95, is the same edge 3.2 defines abstractly, instantiated on real dated nodes.

A period's evidence traces back to the retrieval envelope that supplied its dated observation, per 8: a period's date posterior is only as reliable as the observations feeding its phase model, and linking each observation to an immutable envelope means a later re-fetch cannot silently change a phase model's inputs without a new run id appearing in the model's provenance.

The engine emits three views over the same scored frontier, none a verdict: a per-bin view ranking every hypothesis whose `TIME_BIN` falls in that bin, a per-event view collecting every placement sharing an event's object and place, and a per-pair view collecting every sequence hypothesis over one ordered pair of events, one entry per Allen relation the evidence has touched. Display pruning, a top-N cap on a bin, a collapsed tail on an event, belongs to these views alone; it never touches what 5's generator produced.

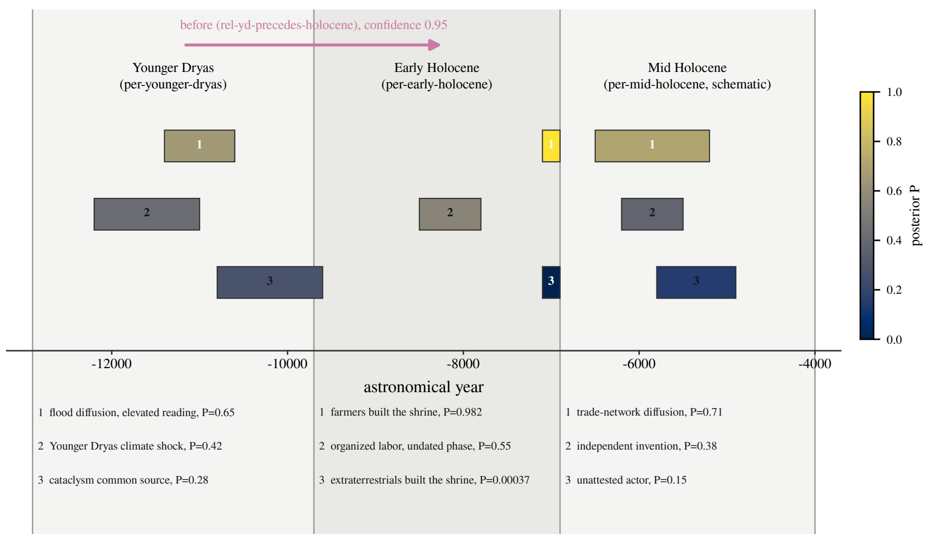

Figure 1: A per-bin timeline export across three time bins on the astronomical-year axis. Each bin holds its ranked placement hypotheses as horizontal bars; bar length is the hypothesis's own interval and bar shade is its projected probability P. The Younger Dryas and Early Holocene bins reuse the exact intervals and P values from the cataclysm-diffusion worked example and the Çatalhöyük farmers-versus-extraterrestrials opinion of 6; the Mid Holocene bin illustrates the same view schematically. The arrow spanning the first two bins is a sequence hypothesis carrying its Allen relation, BEFORE at confidence 0.95.

5 Combinatorial hypothesis space

3.4 and 3.5 fix the two spaces this engine addresses. With vocabulary sizes a working corpus this size can reach, `ACTOR` $= 40$ (thirty human polities plus ten concept-ontology actors, including the five non-consensus actors of 6), `ACTION` $= 50$, `OBJECT` $= 300$, `PLACE` $= 120$, `TIME_BIN` $= 200$ (century bins across a 20,000-year span), `MECHANISM` $= 25$, the two spaces have the following sizes.

Lemma 5.1 (Hypothesis space size)**.** *At the vocabulary sizes stated above,*

$$|H| = 40 \times 50 \times 300 \times 120 \times 200 \times 25 \approx 3.6 \times 10^{11}, \tag{5}$$

$$|H_{\text{seq}}| = 13\,|H|^2 \approx 1.7 \times 10^{24}. \tag{6}$$

Proof. $|H|$ is the product of the six placement slots' vocabulary sizes by 1; substituting the stated sizes gives $40 \times 50 = 2{,}000$, $2{,}000 \times 300 = 600{,}000$, $600{,}000 \times 120 = 72{,}000{,}000$, $72{,}000{,}000 \times 200 = 14{,}400{,}000{,}000$, and $14{,}400{,}000{,}000 \times 25 = 3.6 \times 10^{11}$. $|H_{\text{seq}}|$ is the product of $|H|$, the thirteen-value Allen relation vocabulary of 3.2, and a second copy of $|H|$ by 2; $13 \times (3.6 \times 10^{11})^2 = 13 \times 1.296 \times 10^{23} \approx 1.7 \times 10^{24}$. □

2 plots 5.1's two sizes, 5 and 6, alongside a scale an order of magnitude smaller and one an order of magnitude larger on every axis, showing how fast the address space grows as a corpus names more actors, actions, objects, places, time bins, and mechanisms. No process can materialize a space of 10^{24} points; the engine addresses a point without visiting it.

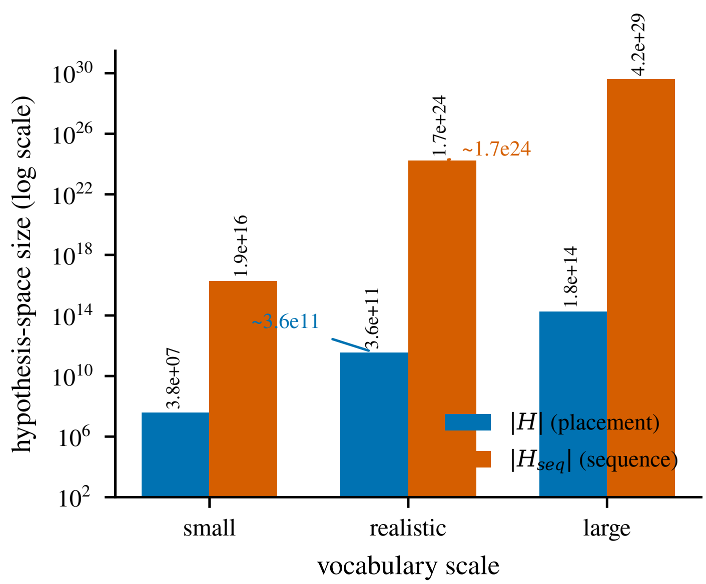

Figure 2: Placement-space and sequence-space size, $|H|$ and $|H_{\text{seq}}|$, at three vocabulary scales on a log axis. The realistic scale is 5.1's own vocabulary sizes; the small and large scales hold every axis an order of magnitude below and above that scale, showing how fast $|H_{\text{seq}}| = 13|H|^2$ grows as the corpus names more actors, actions, objects, places, time bins, and mechanisms.

Lazy enumeration. The generator never iterates H or H_{seq}. For one evidence cluster, it fixes `ACTOR`, `ACTION`, and `MECHANISM` to their full closed vocabularies, exotic actors and mechanisms included, evidence or none, and anchors `OBJECT`, `PLACE`, and `TIME_BIN` to the values that cluster names. The product of those six restricted axes, not H itself, is what one generation pass materializes; the same three anchored axes stay addressable in full through other passes, an era sweep, an object sweep, that call 3 directly on a slot assignment nobody has iterated to. Scoring and display prune the result; generation does not, so a hypothesis at near-zero prior still receives an address and a base rate the moment it is named.

Lemma 5.2 (Address injectivity)**.** *For a fixed slot order and a fixed vocabulary indexing,* address *of 3 is injective: no two distinct slot assignments share an address.*

Proof. Fix a slot assignment slots and its address $n = \prod_k p_k^{v_k+1}$. The primes $p_1, \ldots, p_{13}$ are pairwise distinct by construction, so n's prime factorization is exactly $\{p_k^{v_k+1}\}_k$, with no other prime dividing n and no p_k appearing to any other exponent. The fundamental theorem of arithmetic states this factorization is unique for every positive integer, so recovering v_k for each k, by trial division, dividing by p_k until it fails and recording the exponent minus one, recovers slots exactly. Two distinct slot assignments differ in some v_k, hence differ in the exponent of p_k in their respective factorizations, hence produce distinct n. □

5.2 is why the address needs no lookup table: the alternative, hashing the slot tuple, is fixed-width and URL-friendly but opaque, since no hash inversion recovers the slots without a stored table. The prime encoding recovers them from the integer alone. The `SHA-256` hash is still stored, as `address_hash`, a short display alias the address itself does not need to function.

Definition 5.3 (Neighbor mutation)**.** The neighbor set of a hypothesis h with slot assignment slots is

$$N(h) = \big\{ \operatorname{address}(\text{slots}[k \mapsto c]) \ : \ k \in \text{slots},\ c \in \operatorname{vocab}(k),\ c \neq \text{slots}[k] \big\}, \tag{7}$$

holding every slot but one fixed and ranging that one slot over its vocabulary. $N(h)$ is the evolver's move set: every one-slot mutation of h, addressed by 3 without ever materializing H or H_{seq} around it.

6 Belief model

Evidence groups into nine independent modalities: material or archaeological, textual or documentary, genetic, linguistic, astronomical or radiometric dating, geological and climate, oral tradition and myth, iconographic, and model-based inference. For cluster i of kind κ, the evidence weight is

$$s_i = k(\theta_i) \cdot e_i, \tag{8}$$

with $k(\theta_i)$ the fixed tier weight of 3.7 and e_i the cluster's edge strength. Where system A's original formula folds embedding cosine, fuzzy string match, and shared-motif count into one scalar, $e_i = \min(0.99,\ 0.40\cos + 0.25\,\text{fuz} + 0.10\,\text{motif})$, this pass keeps that formula as one named view, $e_{i,\text{blended-A}}$, and adds separate views $e_{i,\text{semantic}}$, $e_{i,\text{lexical}}$, and $e_{i,\text{motif}}$ for a claim extractor that reports them apart, the way `scientific-discovery` keeps task-family, math, operator, and lexical similarity apart rather than folding structural similarity into one number. A view-aggregation rule, mean, max, or a learned per-view weight, is scoped to its own bead and not decided here; every correlation already scored under the blended formula keeps its stored confidence unchanged.

Within one kind, corroboration gets diminishing returns:

$$D(n) = 1 + \lambda \ln(1+n), \qquad \lambda = 0.5, \tag{9}$$

applied with n replaced by the effective count n_{eff} of 3.10 wherever 9 appears, so two texts that copy one archetype count once rather than twice. Across kinds, let K_+ (respectively K_-) be the number of distinct kinds carrying supporting (refuting) evidence, grouped into three families, material trace, textual trace, and inference, with kind_distance equal to 1 across families and 0 within one. The cross-kind independence bonus is

$$X(K) = 1 + 0.3 \sum_{\text{pairs}} \text{kind\_distance}(\text{pair}), \tag{10}$$

rewarding corroboration from two different kinds over two clusters of one kind. The pooled supporting and refuting weights are $S_+ = X(K_+) \sum_\kappa D(n_{\text{eff},\kappa}) \sum_{i \in \kappa} s_i$ and S_- symmetrically for refuting evidence, with each s_i the per-cluster evidence weight of 8.

Definition 6.1 (Opinion from evidence sums)**.** Fix the total-ignorance weight $W = 2$. Given pooled supporting and refuting weights S_+ and S_- for a hypothesis h, let $r = S_+$ and $s = S_-$ (each already carrying its own 9 discount). The fused opinion is

$$b(h) = \frac{r}{r+s+W}, \qquad d(h) = \frac{s}{r+s+W}, \qquad u(h) = \frac{W}{r+s+W}, \qquad a(h) = \text{sigmoid}\big(L_{\text{prior}}(h)\big), \tag{11}$$

where $L_{\text{prior}}(h) = \sum_{\text{slots}} \ell(c)$ sums the prior logit of every slot's concept, per 3.3. Lean counterpart: `Bucket.Belief.fromEvidence`, proved over its `Rat` counterpart `fromEvidenceQ` (6.2).

Lemma 6.2 (Normalization and probability bound)**.** *For every hypothesis h scored by 11, $b(h) + d(h) + u(h) = 1$ and $P(h) \in [0, 1]$.*

Proof. By 11, $b(h) + d(h) + u(h) = \frac{r+s+W}{r+s+W} = 1$ whenever $r, s \geq 0$ and $W > 0$, both guaranteed by construction ($S_+, S_- \geq 0$ as sums of nonnegative tier weights, and $W = 2$ fixed). For the bound, $P(h) = b(h) + a(h)u(h)$ by 4. Since $a(h) = \text{sigmoid}(\cdot) \in [0, 1]$ and $u(h) \in [0, 1]$ by the normalization just shown, $a(h)u(h) \in [0, u(h)]$, so $P(h) \in [b(h), b(h) + u(h)]$. Since $b(h) + u(h) \leq b(h) + d(h) + u(h) = 1$ and $b(h) \geq 0$, both endpoints lie in $[0, 1]$, hence $P(h) \in [0, 1]$. □

At zero evidence, $r = s = 0$, so $b = d = 0$ and $u = 1$ by 11, giving $P(h) = a(h)$ exactly (Lean: `Bucket.Belief.u_eq_one_of_no_evidence`): a hypothesis nobody has examined reads at its prior, with a stated uncertainty mass of 1, rather than at an unmarked point estimate indistinguishable from a refuted one. Ranking stays compatible with the corpus's existing log-odds signal: with $\Theta_{\text{temporal}}(h)$ the agreement between h's stated interval and its period's date posterior (zero when no date claim is tested), the ranking score

$$L(h) = \ln\left(\frac{P(h)}{1 - P(h)}\right) + \Theta_{\text{temporal}}(h) \tag{12}$$

is a read-out of the opinion rather than the primitive it is built from, so the tournament of 8 keeps working on it unmodified.

Contradiction falls out on its own. The corpus's original truth-score formula subtracted a separate contradiction penalty $\Omega_{\text{contradiction}} = \mu \cdot \min(D(n_+)S_+, D(n_-)S_-)$, with $\mu = 0.5$, whenever both sides carried real weight. 11 needs no such term: when $b(h)$ and $d(h)$ sit close together, $P(h)$ sits close to $a(h)$, since $b = d$ gives $P = (1 - u)/2 + au$, which reduces to a as $u \to 1$ and to $1/2$ only when $a = 1/2$. A contested hypothesis settles at the prior it carries instead of landing at even odds by default, so μ has no counterpart in 11 or 12: the opinion model absorbs what $\Omega_{\text{contradiction}}$ did.

3 plots the opinion triangle, and 4 plots b, d, u, and P against a growing evidence count under 11 and 9, for a fixed tier weight and an uninformative base rate $a = 0.5$: panel (a) accumulates supporting evidence only, panel (b) mixes support and refutation at a 2-to-1 ratio, and both panels show u shrinking as $r + s$ grows regardless of which side leads.

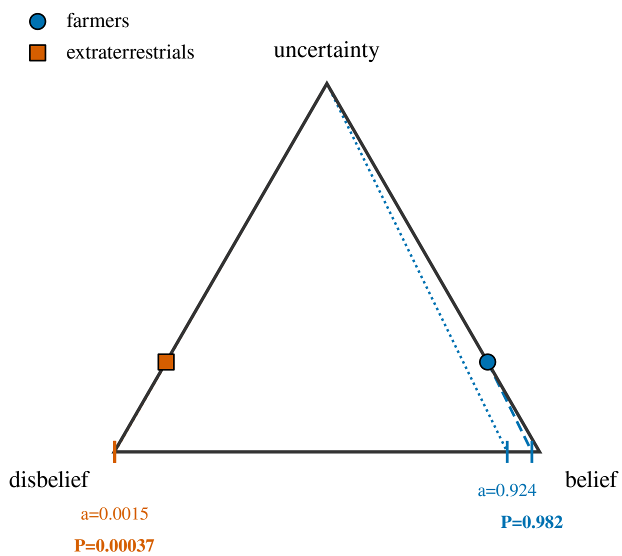

Figure 3: The subjective-logic opinion triangle, vertices belief, disbelief, and uncertainty, with the Çatalhöyük worked example of 2. Each opinion (b, d, u) plots at its barycentric position; the projected probability $P = b + au$ for a base rate a reads off the base edge along the line through the opinion point running parallel to the segment from the uncertainty vertex to the base-rate point $(a, 0)$ on that edge.

Detectability as a likelihood ratio. An absence of evidence is data only when evidence would likely have survived and been found. Detectability (3.11) scales absence evidence in place of a flat tier:

$$e_{\text{absence}} = \delta(\text{period}, \text{medium}, \text{region}) \cdot e_{\text{raw-absence}}, \tag{13}$$

read as the likelihood ratio between observing no evidence when h is true and observing no evidence when h is false: as $\delta \to 0$ that ratio approaches 1 and absence stops discriminating; as $\delta \to 1$ it recovers the flat-tier treatment as its high-detectability limit. A positive find in a low-δ context is the rarer event under this same ratio, so a reviewer can weight it up by the same δ on the corroboration side, by hand rather than by a default rule.

Worked example. Fix `ACTION` = built, `OBJECT` = Çatalhöyük shrine, `PLACE` = Çatalhöyük, `TIME_BIN` = −7000-century, and vary `ACTOR` and `MECHANISM`. `neolithic-farmers-anatolia` (prior logit 2.0) paired with `organized-human-labor` (0.5) gives $L_{\text{prior}} = 2.5$; `extraterrestrials` (−4.0) paired with `unknown- technology` (−2.5) gives $L_{\text{prior}} = -6.5$. Both hypotheses already carry an address and a base rate before either is read against a source. Pool three evidence items against both, flipped by which `ACTOR` they favor: a radiocarbon-dated construction fill with hand-tool marks (material, T1, $e = 0.9$), a tool-mark microwear study (material, T2, $e = 0.5$), and the documentary

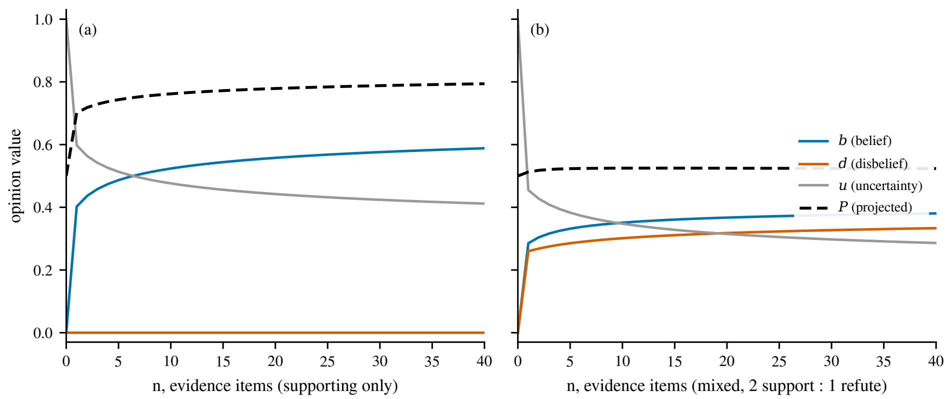

Figure 4: Belief, disbelief, uncertainty, and projected probability against the number of independent evidence items n, for one hypothesis at a fixed tier weight and an uninformative base rate $a = 0.5$. Panel (a) accumulates supporting evidence only; panel (b) mixes support and refutation at a 2-to-1 ratio. Both panels use 11 with $W = 2$ and the diminishing-returns discount of 9 applied to each side before it enters r and s.

silence across every tradition on a non-human builder at the site (textual, T3, $e = 0.6$). All three support farmers and refute the extraterrestrial reading. Within the material kind,

$$\sum k \cdot e = 2.0(0.9) + 1.5(0.5) = 2.55, \quad n_{\text{eff}} = 2, \quad D(2) = 1.549 \Rightarrow 3.951, \tag{14}$$

and within the textual kind,

$$\sum k \cdot e = 1.0(0.6) = 0.60, \quad n_{\text{eff}} = 1, \quad D(1) = 1.347 \Rightarrow 0.808. \tag{15}$$

The pooled weight is $E = 4.759$ across $K = 2$ kinds spanning one cross-family pair, so $X(2) = 1 + 0.3(1) = 1.3$ by 10 and $S_+ = 1.3 \times 4.759 = 6.186$ for the farmers reading, $S_- = 6.186$ for the extraterrestrial reading. 2 carries both readings through 11.

Table 2: Çatalhöyük shrine-construction worked example: the base rate before any evidence is pooled, and the opinion after pooling the three items of 14 and 15.

Quantity	Farmers reading	Extraterrestrial reading
L_{prior}	2.5	−6.5
$a = \text{sigmoid}(L_{\text{prior}})$	0.924	0.0015
r (pooled support)	6.186	0
s (pooled refutation)	0	6.186
b	0.756	0.000
d	0.000	0.756
u	0.244	0.244
$P = b + au$	0.982	0.00037

Both readings land the same direction as the corpus's original sigmoid posterior ($\tau \approx 0.9998$ and $\tau \approx 0.0000031$ under 12 without the opinion layer) and stay separated, but land less extreme on each side: the opinion form keeps 24 percent of the pool as stated uncertainty, so the same three findings move the mainstream reading to 98 percent instead of four nines, and the exotic reading to four parts in ten thousand instead of three parts in a million. The evidence pooled is unchanged; what changed is that the model now states how much of its confidence rests on the prior against how much rests on what was found.

7 Structural unknowns

A gap node (3.12) names an unexcavated site, an untranslated text, an undated stratum, or an unread archive, and the hypotheses reading it would move. Its ranking score folds five factors, borrowed from `scientific-discovery`'s active-retrieval priority rather than resting on expected uncertainty reduction alone:

$$\mathrm{voi}(\mathrm{gap}) = w_1\,\Delta u_{\mathrm{expected}} + w_2\,\mathrm{novelty}(\mathrm{gap}) + w_3\,\mathrm{coverage\_gap}(\mathrm{period}) + w_4\,\mathrm{historical\_gap}(\mathrm{period}) + w_5\,\mathrm{disagreement} \tag{16}$$

with weights $w_1, \ldots, w_5$ starting uniform and calibrated against the holdout runs of 9. Since new evidence always adds to r or s and shrinks u by 11, $\Delta u_{\mathrm{expected}}$ is well defined for every gap without running the retrieval it names; the active-learning queue 16's score orders is the mirror of 9's founder-review queue, one ranks what to judge, the other ranks what to go look at next.

Every slot's vocabulary (3.3) carries a reserved `OTHER` concept with a Dirichlet-process new-concept probability

$$P(\text{new concept in slot}) = \frac{\alpha}{\alpha + N} \tag{17}$$

[21], with N the count of concept instances seen so far in that slot and α a per-slot concentration constant. A hypothesis naming `ACTOR: other-unnamed` still receives an address by 3 and a base rate by 11.

Missing mass (3.13) is estimated by running generation several times under different explore-or-exploit seeds or prior profiles and treating each run's frontier as a sample: f_1/N estimates the unseen share, and the Chao1 richness estimate

$$S_{\mathrm{est}} = S_{\mathrm{obs}} + \frac{f_1^2}{2f_2} \tag{18}$$

[19, 20] sizes the total address space reachable from the evidence on file. The same estimator applies to evidence discovery per period, counting discovery events instead of hypotheses. 5 shows both quantities against a growing number of generation runs, for a simulated generator drawing from a Zipf-distributed pool of 5,000 addressable hypotheses: the Chao1 curve's gap below the pool's known true size after 60 runs of 40 draws each is the caution against reporting a coverage number before the estimator has had enough runs to converge on it.

Coverage share replaces the claim that generation produces the full set. For each time bin,

$$\mathrm{coverage\_share}(\mathrm{bin}) = \frac{\mathrm{hypotheses\_generated\_and\_scored}(\mathrm{bin})}{S_{\mathrm{est}}(\mathrm{bin})}, \tag{19}$$

reported as an interval, ci_low and ci_high, since Chao1 carries an analytic variance. A bin's interval is reported only after it passes a three-part audited-saturation gate, borrowed from `scientific-discovery`: a minimum iteration count, recent normalized novelty below threshold, and stable coverage strata across period, tradition, evidence kind, and rights tier. A bin that has not passed the gate reports a status of generating in place of an interval, so an early, narrow-looking, and misleadingly confident interval never appears before the search has run long enough to earn one.

6 shows one slice of the per-period, per-medium, per-region detectability table 13 scales absence evidence by: rows run from the Upper Paleolithic through the Iron Age and Classical period,

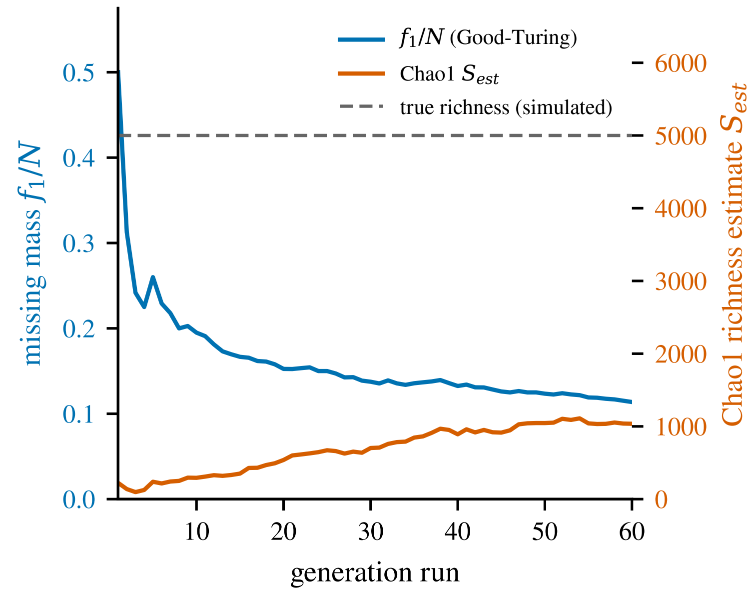

Figure 5: Good-Turing missing mass f_1/N against the number of generation runs, with the Chao1 richness estimate of 18 on a second axis, for a simulated generator drawing from a Zipf-distributed pool of 5,000 addressable hypotheses. The dashed line marks the pool's known true size; the Chao1 curve's gap below that line after 60 runs of 40 draws each is the same caution 9 raises against reporting a coverage number before an estimator has converged.

columns are the nine evidence kinds of 6, and the illustrated values show textual detectability near zero before writing, rising once a period has a literate record, while genetic detectability rises toward the present.

A surprise event is logged when new evidence matches no hypothesis address at any slot-value combination the generator has materialized. Its rate, events per period divided by total evidence discovered that period, is a health metric read two ways: a rising rate says the ontology trails the evidence, a falling rate says either convergence or a generator that has stopped looking, and the two readings are told apart by whether evidence volume that period held steady.

Prior-profile robustness scores one hypothesis under four prior profiles, consensus (L_{prior} as computed today), skeptic (flat or negative priors on every non-attested extraordinary claim), fringe (elevated priors on the exotic actors and mechanisms of 6), and uniform ($L_{\text{prior}} = 0$ everywhere):

$$\text{robustness}(h) = 1 - \Big(\max_p P_p(h) - \min_p P_p(h)\Big), \tag{20}$$

where $P_p(h)$ recomputes 11 and 4 with $a(h)$ read from profile p's L_{prior}. A hypothesis with high robustness(h) earns more trust than one whose rank flips depending on which profile scores it.

8 Engine loop

7 lays out the full loop this engine runs, generate, critique, rank, evolve, review, the shape DeepMind's AI co-scientist runs over biomedical literature [1], with MC-NEST's explicit explore-or-exploit control on the ranking step [9].

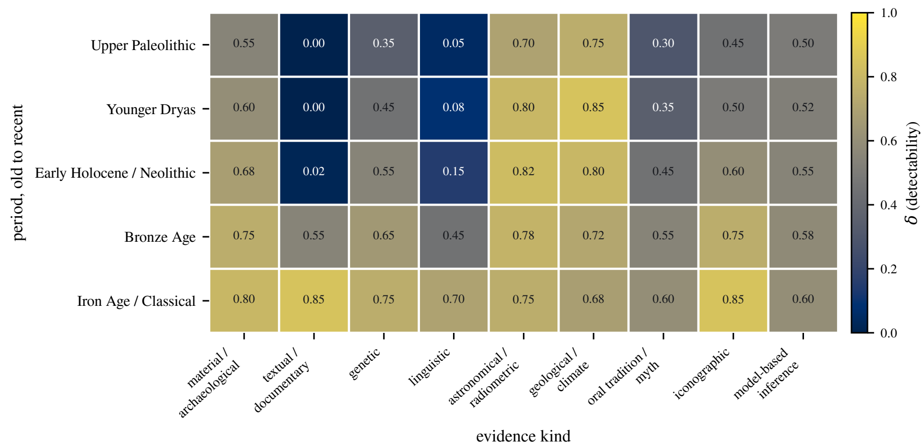

Figure 6: A toy detectability table δ(period, evidence kind), one slice of the per-period, per-medium, per-region table of 3.11. Rows run from the Upper Paleolithic to the Iron Age and Classical period; columns are the nine evidence kinds of 6. Values are illustrative: textual detectability sits near zero before writing and rises once a period has a literate record, while genetic detectability rises toward the present.

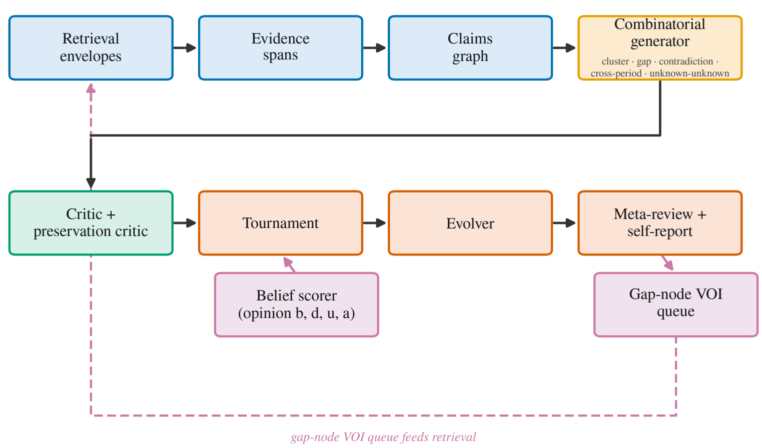

Figure 7: The hypothesis engine's generation loop. Retrieval envelopes ground evidence spans, which populate the claims graph; the combinatorial generator draws placement and sequence hypotheses through four generator kinds, cluster, gap, contradiction, and cross-period, plus the unknown-unknown generator, which proposes slot values outside the current vocabulary. Surviving hypotheses pass through the critic and preservation critic, a belief scorer feeds the ranking tournament, and the evolver recombines, splits, and generalizes the frontier before meta-review and self-report close the run. The gap-node value-of-information queue reads the self-report and routes back into retrieval.

Generator. Four generator kinds feed the frontier: evidence clusters, grouped by embedding, fuzzy match, and shared-motif similarity, enumerate a placement or sequence per 1 or 2; claim gaps yield a hypothesis proposing a missing predicate on an under-specified node; contradictions between two comparable-prior claims yield a hypothesis reconciling or arbitrating them; cross-period analogies over a recurring motif yield a hypothesis testing common cause against independent recurrence. A fifth role, the unknown-unknown generator, proposes only slot values outside the current vocabulary, feeding the open-world mass of 17 directly; it never scores, only proposes. Every hypothesis this stage produces receives an address by 3 before scoring, per 5's lazy-enumeration rule.

Critic and preservation critic. A rule-and-model pass rejects combinations the claims graph itself contradicts, an actor not attested inside the proposed time bin, a place outside every tradition the actor belongs to, before scoring runs. The preservation critic asks, for every surviving hypothesis, what evidence would exist if it were true and whether that evidence could have survived to be found, reading 3.11's table before recommending rejection, so a low detectability score explains an absence before falsity does.

Ranking tournament. Elo is seeded from the posterior, $\mathrm{Elo}_0 = 1500 + 400\,\mathrm{logit}(P(h))$, then updated by pairwise debate rounds. Every fixed number of rounds, 11 recomputes exactly and Elo reseeds from it, keeping the projected probability authoritative over the tournament's running score rather than the reverse.

Evolver. 5.3's one-slot mutation (7) is the evolver's smallest move; it also recombines two hypotheses sharing a cluster into one with a merged claim set, splits an overbroad hypothesis into narrower dated children, and generalizes a set of narrow siblings that agree into one broader periodization claim.

Self-report and target-blind generation. Every run emits a self-report: its assumptions, which slot vocabularies it judged incomplete, its missing-mass estimate (18), and its calibration score against the holdout tests of 9. `scientific-discovery`'s charter states that its retrieval never biases corpus construction toward quantum relevance; this engine generalizes that rule as target-blind generation: generation stays complete and scored the same way regardless of which target reading, orthodox, fringe, or any consensus status between, a hypothesis would end up supporting. A per-run self-report line states whether the generator's proposal rate for non-consensus actor or mechanism values held steady against the run before it; a falling rate under stable evidence volume is the same health signal 7's surprise-rate tracks, read here for target-blindness instead of vocabulary drift. The model's own trained-in sense of a period, cited in 2 as Hist-LLM's near-46-percent recall against Seshat's coded judgments, enters as one evidence kind, `model-prior`, at a low tier, with its own reliability tracked run over run; it never counts as truth on its own.

Target-blind corpus construction and retrieval provenance. Every fetch a mirror job makes is written as an immutable retrieval envelope, linked to a `RetrievalRun` id, so a later re-fetch cannot silently change what an earlier run found. Once a fetch routes through the `x402-research-gateway`, the envelope carries the gateway's manifest fingerprint, protocol version, and citation and lineage counts; until that path exists for history sources, the envelope still carries the provenance shape against the existing corpus mirror job, so the provenance layer needs no rework when the gateway path lands.

9 Calibration and pilot design

Two holdout splits test whether a posterior predicts unseen evidence, run inside a named campaign rather than an ad hoc script call, borrowing `scientific-discovery`'s discipline of a frozen work set sampled across explicit strata (period, tradition, evidence kind, rights tier) before generation starts, with extraction, scoring, and matching failures recorded in the campaign's run log before any repair lands.

Discovery-date holdout. Split each claim's evidence at a cutoff on its recorded discovery date, recompute 11 from pre-cutoff evidence only, then check whether the post-cutoff evidence's support sign matches what the pre-cutoff opinion implied. Score the match rate as a Brier score across every hypothesis with holdout evidence. The first campaign runs this split on a well-dated modern timeline, chosen because its ground truth is settled well enough that a wrong prediction is attributable to the model rather than to the timeline's own uncertainty.

Period holdout. Withhold every claim assigned to one period, generate hypotheses from the remaining periods only, then test the withheld period's real claims against the predicted slot tuples for a match rate. This checks the engine against the claims graph for that period rather than against a trained model's memorized sense of it, the caution the Hist-LLM benchmark raises about a model's memorized sense of a period from training data (2).

Both Brier scores feed a recalibration pass over the tunable constants λ (9) and W (11), fit by grid search and re-run on a schedule as the corpus grows. The corpus's original contradiction penalty μ takes no recalibration pass of its own: 6 folds contradiction into the opinion model directly rather than carrying μ forward as a separate term. Each recalibration carries a run id, the same audit pattern the corpus's mirror jobs already write.

Once the discovery-date holdout has run on a well-dated timeline, the period holdout runs as a frozen campaign against a contested period, chosen from the two candidates in 3. Neither candidate has been run; the table states what each would exercise, not what either would show.

Table 3: Two candidate periods for the first frozen period-holdout campaign, stated as design options. Neither has been run; no result is claimed for either.

Period	Date range	Evidence kinds exercised
Younger Dryas boundary	12.9 to 11.7 ka	Astronomical and radiometric dating, geological and climate, material and archaeological (impact-proxy strata), oral tradition and myth (flood-motif correlations already on file). Detectability is low across every kind, a preliterate period with a thin, contested stratigraphic record.
Neolithic Anatolia	9500 to 7000 BCE	Material and archaeological (Çatalhöyük and comparable sites), genetic (ancient-DNA population continuity), linguistic (proto-language dispersal claims), model-based inference. This paper's own worked example (2) sits inside this window.

10 Limitations

This is a design paper: every claim above is a derivation, checked in 5 and 6 and pointed at its Lean listing in A, and no empirical result is claimed anywhere in it. Several limitations follow directly from that scope.

The model-prior evidence kind (8) mitigates the generating model's own trained-in sense of a period only in part: it lowers that sense to a low-tier, tracked-reliability kind rather than an unaccounted background assumption, but it does not remove the sense itself, and a model whose training corpus underrepresents a tradition still generates fewer hypotheses favoring that tradition's reading before any evidence is read.

The constants $W = 2$ (11) and $\lambda = 0.5$ (9) are uncalibrated: they are the values the corpus's existing truth-score formula already used, kept fixed through the move to an opinion model rather than refit to it. 9's recalibration pass is designed but has not run.

The reserved `OTHER` concept (17) gives a value outside the closed vocabulary an address and a share of the probability mass, without naming that value: a hypothesis routed through `OTHER` carries a base rate and a place in the frontier, leaving the specific actor, action, or mechanism a reader would need to go test it unnamed. Every slot's vocabulary stays closed at the concept level even though its reserved mass is open; naming what `OTHER` stands for on a given run stays a curation step outside 17 itself.

Detectability (3.11) and gap nodes (3.12) have no Lean counterpart in this pass; A names both as pending rather than proving a stand-in for either. The detectability table itself is stated as illustrative in 6: its real values are a claim about the corpus, one new excavation or archive access can revise.

Two gaps sit inside the Lean listings themselves (A). First, `Bucket.Belief.Opinion` is the `Float` shape the running system computes with, and every proved claim, 6.2 included, is proved instead over `Bucket.Belief.OpinionQ`'s `Rat` counterpart, since `Float` carries no Lean order structure a proof can use; nothing in this pass bounds how far a running `Float` computation can drift from its `Rat` proof. Second, `Bucket.Address.encode_injective`, the Lean statement of 5.2's general injectivity claim, is left `sorry`: the pen-and-paper proof in 5 stands on the fundamental theorem of arithmetic, and mechanizing that argument for an arbitrary vocabulary size needs Mathlib's `Nat.factorization` or an equivalent unique-factorization development, unavailable in this Mathlib-free build. Only the bounded case, `encode_injective_bounded`, is proved in full, by exhaustive computation over tuples with every slot index below 2.

The Younger Dryas and Neolithic Anatolia candidates of 3 are both contested periods by design, chosen to exercise different evidence-kind mixes; neither is a validation set with an agreed ground truth, so the period holdout's match rate on either reads against the claims graph on file alone. A campaign run on one candidate does not transfer its calibration to the other without its own holdout pass.

Prior-profile robustness (20) reports a spread across four named profiles chosen by hand; it is not a claim that these four span every reasonable prior an outside reviewer might hold, only that a hypothesis stable across these four has cleared one check a hypothesis stable across zero has not.

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A Lean listings

Every definition in 3 names a Lean counterpart under `lean/Bucket/`, a minimal Lake project (`lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, one `lean_lib` target) built by `make lean`, mirroring `papers/template/lean/`. All six files below are on disk and `lake build` passes over every one of them; the guard on each listing below is a defensive fallback rather than a live condition.

Bucket.Timeline (3.1, 3.2).

```
/-!
# Timeline substrate: intervals and Allen's interval algebra

`TIMELINE-AND-COMBINATORICS-SPEC.md` Section 1 fixes the engine's time axis as a
signed integer line in astronomical years and names Allen's thirteen
interval relations as the relation vocabulary for every sequence
hypothesis. This file gives both a Lean 4 home: `Interval` carries the
axis and an uncertainty tag, `AllenRelation` is the thirteen-constructor
relation type, and `relate` decides which relation holds between two
intervals.

Builds in core Lean 4 plus the `omega` and `decide` tactics that ship with
it; no Mathlib dependency (`papers/PAPER-STANDARDS.md`, "Lean rule"). The
converse proofs unfold a thirteen-branch decision chain and need more
than the default elaboration budget.
-/

set_option maxHeartbeats 4000000

namespace Bucket.Timeline

/-- How an interval's boundary is known, reusing the four-way split
`TIMELINE-AND-COMBINATORICS-SPEC.md` Section 1's `uncertainty.distribution` field
draws (`point`, `uniform`, `normal`, `oxcal-posterior`). `sampled` stands
in for the spec's `oxcal-posterior` case, whose payload is a sampled
posterior curve rather than a closed-form distribution. -/
inductive Uncertainty where
  | point
  | uniform
  | normal
  | sampled
  deriving DecidableEq, Repr, BEq

/-- A dated interval on the engine's astronomical-year axis
```

```lean
(`TIMELINE-AND-COMBINATORICS-SPEC.md` Section 1). `start <= stop` is carried as a
proof field rather than assumed, so no downstream lemma about the interval
has to re-derive it. -/
structure Interval where
  start : Int
  stop : Int
  le : start <= stop
  uncertainty : Uncertainty := .point
  deriving Repr

/-- Allen's thirteen interval relations
(`TIMELINE-AND-COMBINATORICS-SPEC.md` Section 1), the relation vocabulary for
every sequence hypothesis and every edge between two dated nodes. -/
inductive AllenRelation where
  | before
  | after
  | meets
  | metBy
  | overlaps
  | overlappedBy
  | starts
  | startedBy
  | during
  | contains
  | finishes
  | finishedBy
  | equal
  deriving DecidableEq, Repr, BEq

open AllenRelation in
/-- Decide which of the thirteen Allen relations holds between `a` and
`b`. The thirteen conditions below are checked in a fixed order, each
`else` falling through to the next, so exactly one branch fires for any
pair of intervals: `relate_total` below is the formal statement that this
is a well-defined function (trivially, since every Lean function is
total), and `relate_before_iff` through `relate_metBy_iff` characterize
each of the first four branches in closed arithmetic form. -/
def relate (a b : Interval) : AllenRelation :=
  if a.stop < b.start then before
  else if b.stop < a.start then after
  else if a.stop = b.start then meets
  else if b.stop = a.start then metBy
  else if a.start = b.start /\ a.stop = b.stop then equal
  else if a.start = b.start /\ a.stop < b.stop then starts
  else if a.start = b.start then startedBy
  else if a.stop = b.stop /\ a.start < b.start then finishedBy
  else if a.stop = b.stop then finishes
  else if b.start < a.start /\ a.stop < b.stop then during
  else if a.start < b.start /\ b.stop < a.stop then contains
  else if a.start < b.start then overlaps
```

```
  else overlappedBy

/-- ‘relate‘ is total: every pair of intervals gets exactly one relation
    .
This holds by construction, since ‘relate‘ is a Lean function (every
function returns exactly one value for each input); the existence half
and the uniqueness half are spelled out separately here rather than
folded into ‘exists !‘ notation, which needs Mathlib. -/
theorem relate_total (a b : Interval) :
    exists  r : AllenRelation, relate a b = r /\ forall  r’, relate a b
       = r’ -> r’ = r :=
  <relate a b, rfl, fun _ hr’ => hr’.symm>

theorem relate_before_iff (a b : Interval) :
    relate a b = .before <-> a.stop < b.start := by
  have := a.le; have := b.le
  unfold relate
  constructor
  . intro h
    iterate 40 (all_goals (try split at h))
    all_goals (try (exact absurd h (by decide)))
    all_goals omega
  . intro h
    rw [if_pos h]

theorem relate_after_iff (a b : Interval) :
    relate a b = .after <-> b.stop < a.start := by
  have := a.le; have := b.le
  unfold relate
  constructor
  . intro h
    iterate 40 (all_goals (try split at h))
    all_goals (try (exact absurd h (by decide)))
    all_goals omega
  . intro h
    have h1 : not  (a.stop < b.start) := by omega
    rw [if_neg h1, if_pos h]

/-- ‘relate a b = before‘ and ‘relate b a = after‘ are the same fact
    seen
from either interval, with no side condition: a strict gap between two
intervals reads as ‘before‘ from the earlier one and ‘after‘ from the
later one, unconditionally. -/
theorem relate_converse_before_after (a b : Interval) :
    relate a b = .before <-> relate b a = .after := by
  rw [relate_before_iff, relate_after_iff]

theorem relate_meets_iff (a b : Interval) :
    relate a b = .meets <->
      (not  a.stop < b.start) /\ (not  b.stop < a.start) /\ a.stop = b.
         start := by
  have hia := a.le; have hib := b.le
  constructor
  . intro h
```

```
    have h1 : not  a.stop < b.start := by
      intro hc
      have hh := (relate_before_iff a b).mpr hc
      rw [h] at hh
      exact absurd hh (by decide)
    have h2 : not  b.stop < a.start := by
      intro hc
      have hh := (relate_after_iff a b).mpr hc
      rw [h] at hh
      exact absurd hh (by decide)
    refine <h1, h2, ?_>
    unfold relate at h
    rw [if_neg h1, if_neg h2] at h
    by_cases h3 : a.stop = b.start
    . exact h3
    . exfalso
      rw [if_neg h3] at h
      by_cases h4 : b.stop = a.start
      . rw [if_pos h4] at h; exact absurd h (by decide)
      . rw [if_neg h4] at h
        by_cases h5 : a.start = b.start /\ a.stop = b.stop
        . rw [if_pos h5] at h; exact absurd h (by decide)
        . rw [if_neg h5] at h
          by_cases h6 : a.start = b.start /\ a.stop < b.stop
          . rw [if_pos h6] at h; exact absurd h (by decide)
          . rw [if_neg h6] at h
            by_cases h7 : a.start = b.start
            . rw [if_pos h7] at h; exact absurd h (by decide)
            . rw [if_neg h7] at h
              by_cases h8 : a.stop = b.stop /\ a.start < b.start
              . rw [if_pos h8] at h; exact absurd h (by decide)
              . rw [if_neg h8] at h
                by_cases h9 : a.stop = b.stop
                . rw [if_pos h9] at h; exact absurd h (by decide)
                . rw [if_neg h9] at h
                  by_cases h10 : b.start < a.start /\ a.stop < b.stop
                  . rw [if_pos h10] at h; exact absurd h (by decide)
                  . rw [if_neg h10] at h
                    by_cases h11 : a.start < b.start /\ b.stop < a.stop
                    . rw [if_pos h11] at h; exact absurd h (by decide)
                    . rw [if_neg h11] at h
                      by_cases h12 : a.start < b.start
                      . rw [if_pos h12] at h; exact absurd h (by decide)
                      . rw [if_neg h12] at h
                        exact absurd h (by decide)
  . rintro <h1, h2, h3>
    unfold relate
    rw [if_neg h1, if_neg h2, if_pos h3]

theorem relate_metBy_iff (a b : Interval) :
    relate a b = .metBy <->
      (not  a.stop < b.start) /\ (not  b.stop < a.start) /\
        (not  a.stop = b.start) /\ b.stop = a.start := by
  have hia := a.le; have hib := b.le
```

```
  constructor
  . intro h
    have h1 : not  a.stop < b.start := by
      intro hc
      have hh := (relate_before_iff a b).mpr hc
      rw [h] at hh; exact absurd hh (by decide)
    have h2 : not  b.stop < a.start := by
      intro hc
      have hh := (relate_after_iff a b).mpr hc
      rw [h] at hh; exact absurd hh (by decide)
    have h3 : not  a.stop = b.start := by
      intro hc
      have hh := (relate_meets_iff a b).mpr <h1, h2, hc>
      rw [h] at hh; exact absurd hh (by decide)
    refine <h1, h2, h3, ?_>
    unfold relate at h
    rw [if_neg h1, if_neg h2, if_neg h3] at h
    by_cases h4 : b.stop = a.start
    . exact h4
    . exfalso
      rw [if_neg h4] at h
      by_cases h5 : a.start = b.start /\ a.stop = b.stop
      . rw [if_pos h5] at h; exact absurd h (by decide)
      . rw [if_neg h5] at h
        by_cases h6 : a.start = b.start /\ a.stop < b.stop
        . rw [if_pos h6] at h; exact absurd h (by decide)
        . rw [if_neg h6] at h
          by_cases h7 : a.start = b.start
          . rw [if_pos h7] at h; exact absurd h (by decide)
          . rw [if_neg h7] at h
            by_cases h8 : a.stop = b.stop /\ a.start < b.start
            . rw [if_pos h8] at h; exact absurd h (by decide)
            . rw [if_neg h8] at h
              by_cases h9 : a.stop = b.stop
              . rw [if_pos h9] at h; exact absurd h (by decide)
              . rw [if_neg h9] at h
                by_cases h10 : b.start < a.start /\ a.stop < b.stop
                . rw [if_pos h10] at h; exact absurd h (by decide)
                . rw [if_neg h10] at h
                  by_cases h11 : a.start < b.start /\ b.stop < a.stop
                  . rw [if_pos h11] at h; exact absurd h (by decide)
                  . rw [if_neg h11] at h
                    by_cases h12 : a.start < b.start
                    . rw [if_pos h12] at h; exact absurd h (by decide)
                    . rw [if_neg h12] at h
                      exact absurd h (by decide)
  . rintro <h1, h2, h3, h4>
    unfold relate
    rw [if_neg h1, if_neg h2, if_neg h3, if_pos h4]

/-- `relate a b = meets` and `relate b a = metBy` agree, with one
non-degeneracy side condition. `meets` and `metBy` share a boundary
    check
(`a.stop = b.start` versus `b.stop = a.start`) that both hold at once
```

```
exactly when `a` and `b` collapse to the same zero-length instant
(`a.start = a.stop = b.start = b.stop`); `relate` breaks that tie in
`meets`'s favor on both sides, since `meets` is checked first in the
definition above, so the pure converse fails on that one instant-on-
    instant
case. Excluding it (at least one interval has positive length) restores
the converse. -/
theorem relate_converse_meets_metBy (a b : Interval)
    (hnondeg : a.start < a.stop \/ b.start < b.stop) :
    relate a b = .meets <-> relate b a = .metBy := by
  have hia := a.le; have hib := b.le
  rw [relate_meets_iff, relate_metBy_iff]
  constructor
  . rintro <h1, h2, h3>
    exact <by omega, by omega, by omega, by omega>
  . rintro <h1, h2, h3, h4>
    exact <by omega, by omega, by omega>

end Bucket.Timeline
```

Bucket.Concept (3.3).

```
/-!
# Concept ontology: slots, concepts, and the open-world vocabulary

`TIMELINE-AND-COMBINATORICS-SPEC.md` Section 2 gives a hypothesis its
    slot frame,
ACTOR/ACTION/OBJECT/PLACE/TIME/MECHANISM/RELATION, and a `concept` node
    per
slot value carrying its own editable base rate. `IDEAL-STATE-AND-
    UNKNOWNS-
SPEC.md` Section 6a adds an open-world `OTHER` concept to every slot so
    a value
outside the closed vocabulary can still be addressed. This file gives
    both
a Lean home.
-/

namespace Bucket.Concept

/-- The seven hypothesis slots (`TIMELINE-AND-COMBINATORICS-SPEC.md`
    Section 2).
The first six fill a placement hypothesis; `relation` fills a sequence
hypothesis only, drawn from `Bucket.Timeline.AllenRelation`. -/
inductive Slot where
  | actor
  | action
  | object
  | place
  | time
  | mechanism
  | relation
  deriving DecidableEq, Repr, BEq
```

```
/-- How a concept stands against the corpus's own reading, extending the
`stance` vocabulary `ENTITY-MODEL.md` Section 5 already uses on claims
(`TIMELINE-AND-COMBINATORICS-SPEC.md` Section 2's `consensus_status`). `
    other`
marks the open-world placeholder from `IDEAL-STATE-AND-UNKNOWNS-SPEC.md`
Section 6a rather than a graded stance. -/
inductive ConsensusStatus where
  | consensus
  | contested
  | fringe
  | other
  deriving DecidableEq, Repr, BEq

/-- A single slot value: an id, the slot it fills, an editable base rate
in log-odds (`prior_logit`), and a consensus reading
(`TIMELINE-AND-COMBINATORICS-SPEC.md` Section 2). `priorLogit` is a `
    Float` here
to match the field's use as a tunable, human-edited number; `Bucket.
    Belief`
carries the `Rat` counterpart wherever a theorem needs to compute with
    it. -/
structure Concept where
  id : String
  slot : Slot
  priorLogit : Float
  consensusStatus : ConsensusStatus
  deriving Repr

/-- The canonical id of a slot's open-world placeholder concept. -/
def otherId : Slot -> String
  | .actor => "other-actor"
  | .action => "other-action"
  | .object => "other-object"
  | .place => "other-place"
  | .time => "other-time"
  | .mechanism => "other-mechanism"
  | .relation => "other-relation"

/-- The open-world placeholder concept for a slot
(`IDEAL-STATE-AND-UNKNOWNS-SPEC.md` Section 6a): a near-neutral base
    rate
(`priorLogit := 0`) and `consensusStatus := .other`, so a value outside
the closed vocabulary still gets an address, a prior, and a place in the
frontier. -/
def otherConcept (s : Slot) : Concept :=
  { id := otherId s, slot := s, priorLogit := 0.0, consensusStatus := .
      other }

/-- A slot-indexed vocabulary that is never missing its open-world
placeholder: `hasOther` is a proof field, not a claim to check later, so
every `Vocabulary` value carries the guarantee by construction. -/
structure Vocabulary where
  bySlot : Slot -> List Concept
```

```
  hasOther : forall  s : Slot, otherConcept s in bySlot s

/-- Looking up the ‘other‘ concept in any slot always succeeds: it is a
member of that slot’s vocabulary list for every vocabulary and every
    slot,
by ‘Vocabulary.hasOther‘. This is the "lookup of ‘other‘ succeeds for
every slot" guarantee ‘IDEAL-STATE-AND-UNKNOWNS-SPEC.md‘ Section 6a asks
    for,
stated as its own theorem so callers do not have to know the field name
    to
rely on it. -/
theorem lookup_other_succeeds (v : Vocabulary) (s : Slot) :
    otherConcept s in v.bySlot s :=
  v.hasOther s

end Bucket.Concept
```

Bucket.Hypothesis (3.4, 3.5).

```
import Bucket.Timeline
import Bucket.Concept

/-!
# Hypotheses: placements and sequences

‘TIMELINE-AND-COMBINATORICS-SPEC.md‘ intro paragraph: "Every hypothesis
takes one of two base forms: a placement, ’event E happened in time
interval T,’ or a sequence, ’E1 then E2.’" A placement fills the six
non-relational slots from ‘Bucket.Concept.Slot‘ and pins them to a
‘Bucket.Timeline.Interval‘; a sequence pairs two placements with the
Allen relation between them.
-/

namespace Bucket.Hypothesis

/-- A placement hypothesis: "actor performed action on object at place
via mechanism, during this interval"
(‘TIMELINE-AND-COMBINATORICS-SPEC.md‘ Section 2-3). Slot values are
    carried by
‘Bucket.Concept.Concept.id‘ rather than the concept itself, so a
placement is a plain data record independent of which ‘Vocabulary‘ its
concepts happen to live in. -/
structure Placement where
  actor : String
  action : String
  object : String
  place : String
  mechanism : String
  interval : Bucket.Timeline.Interval
  deriving Repr

/-- A sequence hypothesis: two placements related by one of Allen’s
thirteen relations (‘TIMELINE-AND-COMBINATORICS-SPEC.md‘ Section 2-3,
```

```
`H_seq = H x ALLEN_RELATION x H`). -/
structure Sequence where
  first : Placement
  second : Placement
  relation : Bucket.Timeline.AllenRelation
  deriving Repr

end Bucket.Hypothesis
```

Bucket.Address (3.6, 5.2).

```
/-!
# Godel prime encoding for hypothesis addresses

`TIMELINE-AND-COMBINATORICS-SPEC.md` Section 3 fixes the first six
    primes
(2, 3, 5, 7, 11, 13) to the six placement slots in order and addresses a
hypothesis as the product of `prime_slot ^ (vocab_index + 1)` over its
slots. The fundamental theorem of arithmetic (every natural number has
exactly one prime factorization) is why two hypotheses with distinct
slot-index tuples never share an address: the exponent of each fixed
prime in the product recovers that slot's index uniquely.

The general injectivity theorem below needs Mathlib's `Nat.factorization
    `
(or an equivalent unique-factorization development) to go through for an
arbitrary vocabulary size; that is unavailable in this Mathlib-free
    build
(`papers/PAPER-STANDARDS.md`, "Lean rule": depend on Mathlib only when a
lemma needs it, and this project stays in core Lean to keep `lake build`
network-free). `encode_injective` is stated and left `sorry`, with a
    TODO
naming exactly what closes it. `encode_injective_bounded` proves the
    same
injectivity for small, fixed vocabulary sizes in full, by direct
computation, standing in for the general result until Mathlib is
available.
-/

namespace Bucket.Address

/-- A hypothesis's six-slot vocabulary-index tuple, in
ACTOR/ACTION/OBJECT/PLACE/TIME_BIN/MECHANISM order
(`TIMELINE-AND-COMBINATORICS-SPEC.md` Section 3). Each field is the slot
    's
vocabulary index, per an injective, per-slot `index : concept id -> Nat`
assignment (`vocab_index` in the spec) that this structure's caller is
responsible for constructing; `SlotTuple` only carries the resulting
numbers. -/
structure SlotTuple where
  actor : Nat
  action : Nat
  object : Nat
```

```
  place : Nat
  timeBin : Nat
  mechanism : Nat
  deriving DecidableEq, Repr

/-- The Godel address of a slot-index tuple: the product of each slot's
fixed prime raised to the tuple's index at that slot, plus one
(`TIMELINE-AND-COMBINATORICS-SPEC.md` Section 3, `address(slots)`). -/
def encode (t : SlotTuple) : Nat :=
  2 ^ (t.actor + 1) * 3 ^ (t.action + 1) * 5 ^ (t.object + 1) *
    7 ^ (t.place + 1) * 11 ^ (t.timeBin + 1) * 13 ^ (t.mechanism + 1)

/-- Distinct slot tuples give distinct addresses: `encode` never
    collides.
TODO: this needs unique prime factorization (Mathlib's `Nat.
    factorization`
or equivalent) to prove for arbitrary tuples, since the argument is "the
exponent of 2 in `encode t` is `t.actor + 1`, of 3 is `t.action + 1`,
    and
so on, by uniqueness of factorization, so equal addresses force equal
exponents force equal tuples." That argument is not available without
Mathlib in this build; see `encode_injective_bounded` below for the same
fact proved directly for small tuples. -/
theorem encode_injective : Function.Injective encode := by
  sorry

/-- The bounded case, proved in full: two tuples whose every field stays
below 2 (so every exponent stays below 3) collide only when equal.
Settled by exhaustive computation over the 2^6 x 2^6 grid of tuples, no
factorization theory needed. This is the concrete fact
`encode_injective`'s general statement reduces to once the vocabulary
size at every slot is fixed and small, and is the load-bearing sanity
check that the encoding scheme is sound before the general theorem is
available. -/
theorem encode_injective_bounded
    (a1 b1 c1 d1 e1 f1 a2 b2 c2 d2 e2 f2 : Fin 2)
    (h : 2 ^ (a1.val + 1) * 3 ^ (b1.val + 1) * 5 ^ (c1.val + 1) *
           7 ^ (d1.val + 1) * 11 ^ (e1.val + 1) * 13 ^ (f1.val + 1) =
         2 ^ (a2.val + 1) * 3 ^ (b2.val + 1) * 5 ^ (c2.val + 1) *
           7 ^ (d2.val + 1) * 11 ^ (e2.val + 1) * 13 ^ (f2.val + 1)) :
    a1 = a2 /\ b1 = b2 /\ c1 = c2 /\ d1 = d2 /\ e1 = e2 /\ f1 = f2 := by
  revert h
  revert a1 b1 c1 d1 e1 f1 a2 b2 c2 d2 e2 f2
  decide

end Bucket.Address
```

Bucket.Belief (3.8, 3.9, 6.1, 6.2).

```
/-!
# Belief as a subjective-logic opinion

`IDEAL-STATE-AND-UNKNOWNS-SPEC.md` Section 2 replaces the single truth
```

```
    score
`tau` with a subjective-logic opinion `(b, d, u, a)`: belief, disbelief,
uncertainty mass, and base rate, with `b + d + u = 1` and projected
probability `P = b + a.u`. `Opinion` below is the `Float` shape a
    running
system computes with; `Float` has no usable, provable algebra in Lean (
    no
`LinearOrder`, no `decide`-able equality that respects arithmetic), so
    no
theorem in this file is stated about it. `OpinionQ` carries the same
shape over `Rat`, where every theorem below lives.
-/

namespace Bucket.Belief

/-- A subjective-logic opinion over `Float`
(`IDEAL-STATE-AND-UNKNOWNS-SPEC.md` Section 2): belief, disbelief,
    uncertainty
mass, and base rate. This is the shape a running system computes with;
    see
`OpinionQ` below for the provable counterpart. -/
structure Opinion where
  b : Float
  d : Float
  u : Float
  a : Float
  deriving Repr

/-- Build an opinion from pooled supporting evidence weight `r`, pooled
refuting evidence weight `s`, the total-ignorance weight `W` (subjective
logic's standard constant, `W = 2` in `IDEAL-STATE-AND-UNKNOWNS-SPEC.md`
Section 2), and a base rate `a` computed upstream as `sigmoid(L_prior(h)
    )`. -/
def fromEvidence (r s W a : Float) : Opinion :=
  { b := r / (r + s + W), d := s / (r + s + W), u := W / (r + s + W), a
      := a }

/-- The projected probability `P(h) = b + a.u`
(`IDEAL-STATE-AND-UNKNOWNS-SPEC.md` Section 2). -/
def project (omega : Opinion) : Float :=
  omega.b + omega.a * omega.u

/-- Cumulative fusion of two independent opinions sharing one base rate
(Josang's subjective logic; `IDEAL-STATE-AND-UNKNOWNS-SPEC.md` Section 3
     pools
evidence the same way when it moves from raw counts to `n_eff`). The
denominator `u1 + u2 - u1*u2` vanishes only when both opinions are fully
dogmatic (`u1 = u2 = 0`); that case returns the neutral opinion
(`u := 1`) rather than dividing by zero, since cumulative fusion has no
standard reading for two opinions that both claim total certainty. -/
def fuse (omega1 omega2 : Opinion) : Opinion :=
  let denom := omega1.u + omega2.u - omega1.u * omega2.u
  if denom == 0 then
    { b := 0, d := 0, u := 1, a := omega1.a }
```

```
  else
    { b := (omega1.b * omega2.u + omega2.b * omega1.u) / denom
      d := (omega1.d * omega2.u + omega2.d * omega1.u) / denom
      u := (omega1.u * omega2.u) / denom
      a :=
        let adenom := omega1.u + omega2.u - 2 * omega1.u * omega2.u
        if adenom == 0 then (omega1.a + omega2.a) / 2
        else (omega1.a * omega2.u + omega2.a * omega1.u - (omega1.a +
           omega2.a) * omega1.u * omega2.u) / adenom }

/-- The `Rat` counterpart of `Opinion`, where every claim below is
proved. -/
structure OpinionQ where
  b : Rat
  d : Rat
  u : Rat
  a : Rat
  deriving Repr, DecidableEq

def fromEvidenceQ (r s W a : Rat) : OpinionQ :=
  { b := r / (r + s + W), d := s / (r + s + W), u := W / (r + s + W), a
     := a }

def projectQ (omega : OpinionQ) : Rat :=
  omega.b + omega.a * omega.u

/-- A nonnegative numerator over a positive denominator is nonnegative.
   -/
private theorem div_nonneg' {x y : Rat} (hx : 0 <= x) (hy : 0 < y) : 0
   <= x / y := by
  rw [Rat.div_def]
  exact Rat.mul_nonneg hx (Rat.le_of_lt (Rat.inv_pos.mpr hy))

/-- A numerator bounded by a positive denominator divides to at most 1.
   -/
private theorem div_le_one' {x y : Rat} (hxy : x <= y) (hy : 0 < y) : x
   / y <= 1 := by
  rw [Rat.div_def]
  have hinv : 0 <= y^-1 := Rat.le_of_lt (Rat.inv_pos.mpr hy)
  calc x * y^-1 <= y * y^-1 := Rat.mul_le_mul_of_nonneg_right hxy hinv
    _ = 1 := by rw [<- Rat.div_def]; grind

/-- `fromEvidenceQ`'s three masses sum to one, whenever the evidence
weights are nonnegative and the ignorance weight is positive
(`IDEAL-STATE-AND-UNKNOWNS-SPEC.md` Section 2). -/
theorem sum_eq_one (r s W a : Rat) (hr : 0 <= r) (hs : 0 <= s) (hW : 0 <
    W) :
    (fromEvidenceQ r s W a).b + (fromEvidenceQ r s W a).d +
      (fromEvidenceQ r s W a).u = 1 := by
  show r / (r + s + W) + s / (r + s + W) + W / (r + s + W) = 1
  grind

/-- The projected probability of a well-formed opinion (nonnegative
belief and disbelief, masses summing to one) stays in `[0, 1]` whenever
```

```
the base rate does. `b`, `d`, `u` are not free here: an opinion whose
masses do not sum to one or that carries a negative mass is not one
`fromEvidenceQ`, or subjective logic, would ever produce. -/
theorem project_mem_unit (omega : OpinionQ)
    (hb : 0 <= omega.b) (hd : 0 <= omega.d) (hu : 0 <= omega.u) (hsum :
        omega.b + omega.d + omega.u = 1)
    (ha0 : 0 <= omega.a) (ha1 : omega.a <= 1) :
    0 <= projectQ omega /\ projectQ omega <= 1 := by
  show 0 <= omega.b + omega.a * omega.u /\ omega.b + omega.a * omega.u
      <= 1
  have hau0 : 0 <= omega.a * omega.u := Rat.mul_nonneg ha0 hu
  have hau1 : omega.a * omega.u <= 1 * omega.u := Rat.
      mul_le_mul_of_nonneg_right ha1 hu
  grind

/-- With no evidence on either side, an opinion carries full uncertainty
mass and its projection is exactly its base rate: this is
`IDEAL-STATE-AND-UNKNOWNS-SPEC.md` Section 2's reading of a hypothesis
    nobody has
looked at yet, "the opinion form adds one missing tag: that number
    stands
in for evidence, and is not evidence itself." -/
theorem u_eq_one_of_no_evidence (W a : Rat) (hW : 0 < W) :
    (fromEvidenceQ 0 0 W a).u = 1 /\ projectQ (fromEvidenceQ 0 0 W a) =
        a := by
  constructor
  . show W / (0 + 0 + W) = 1
    grind
  . show (0 : Rat) / (0 + 0 + W) + a * (W / (0 + 0 + W)) = a
    grind

end Bucket.Belief
```

Bucket.Unknowns (3.13).

```
/-!
# Missing-mass estimation: Good-Turing and Chao1

`IDEAL-STATE-AND-UNKNOWNS-SPEC.md` Section 6b: "Good-Turing estimates
    the unseen
share of the hypothesis space as `f1/N`... Chao1 estimates total
richness: `S_est = S_obs + f1^2 / (2 * f2)`," with `f2 = 0` handled by
    the
fallback `S_obs + f1(f1-1)/2`.
-/

namespace Bucket.Unknowns

/-- The Good-Turing missing-mass estimate: the share of the hypothesis
space made of things seen exactly once, `n1 / N`
(`IDEAL-STATE-AND-UNKNOWNS-SPEC.md` Section 6b). `Rat`'s division
    convention
(`x / 0 = 0`) makes `missingMass n1 0 = 0`, matching an empty corpus
```

```
having no missing mass to speak of. -/
def missingMass (n1 N : Nat) : Rat :=
  (n1 : Rat) / (N : Rat)

/-- The Chao1 richness estimate over observed species count `sObs`,
singleton count `f1`, and doubleton count `f2`
(`IDEAL-STATE-AND-UNKNOWNS-SPEC.md` Section 6b). The `f2 = 0` case falls
     back to
`sObs + f1(f1-1)/2`; the subtraction `f1 - 1` happens in `Nat` (
    truncating
to `0` when `f1 = 0`) before the cast to `Rat`, so the fallback term
    never
goes negative. -/
def chao1 (sObs f1 f2 : Nat) : Rat :=
  if f2 = 0 then
    (sObs : Rat) + ((f1 * (f1 - 1) : Nat) : Rat) / 2
  else
    (sObs : Rat) + ((f1 ^ 2 : Nat) : Rat) / (2 * (f2 : Rat))

private theorem div_nonneg' {x y : Rat} (hx : 0 <= x) (hy : 0 < y) : 0
    <= x / y := by
  rw [Rat.div_def]
  exact Rat.mul_nonneg hx (Rat.le_of_lt (Rat.inv_pos.mpr hy))

private theorem div_le_one' {x y : Rat} (hxy : x <= y) (hy : 0 < y) : x
    / y <= 1 := by
  rw [Rat.div_def]
  have hinv : 0 <= y^-1 := Rat.le_of_lt (Rat.inv_pos.mpr hy)
  calc x * y^-1 <= y * y^-1 := Rat.mul_le_mul_of_nonneg_right hxy hinv
    _ = 1 := by rw [<- Rat.div_def]; grind

/-- The missing-mass estimate never exceeds one, whenever the singleton
count is at most the total count (as it always is: `n1` counts a subset
of the `N` observations). -/
theorem missingMass_le_one (n1 N : Nat) (h : n1 <= N) : missingMass n1 N
     <= 1 := by
  show (n1 : Rat) / (N : Rat) <= 1
  by_cases hN : N = 0
  . have hn1 : n1 = 0 := by omega
    subst hN; subst hn1
    grind
  . have hNpos : 0 < N := Nat.pos_of_ne_zero hN
    have hNpos' : (0 : Rat) < (N : Rat) := by exact_mod_cast hNpos
    have hle : (n1 : Rat) <= (N : Rat) := by exact_mod_cast h
    exact div_le_one' hle hNpos'

/-- Chao1 never estimates less richness than what was directly observed:
the correction term is always nonnegative, in both the `f2 = 0` fallback
and the general case. -/
theorem chao1_ge_sObs (sObs f1 f2 : Nat) : (sObs : Rat) <= chao1 sObs f1
     f2 := by
  show (sObs : Rat) <= chao1 sObs f1 f2
  unfold chao1
  split
```

```
  . have : (0 : Rat) <= ((f1 * (f1 - 1) : Nat) : Rat) / 2 :=
      div_nonneg' (by exact_mod_cast Nat.zero_le _) (by decide)
    grind
  . rename_i hf2
    have hf2pos : 0 < f2 := Nat.pos_of_ne_zero hf2
    have h2f2pos : (0 : Rat) < 2 * (f2 : Rat) := by
      have : (0 : Rat) < (f2 : Rat) := by exact_mod_cast hf2pos
      grind
    have : (0 : Rat) <= ((f1 ^ 2 : Nat) : Rat) / (2 * (f2 : Rat)) :=
      div_nonneg' (by exact_mod_cast Nat.zero_le _) h2f2pos
    grind

end Bucket.Unknowns
```

`Bucket.Address.encode_injective`, the Lean statement of 5.2's general injectivity claim, is the one result this paper needs machine-checked and leaves `sorry`, per 10; `Bucket.Belief` discharges 6.2 in full, as `sum_eq_one` and `project_mem_unit` over the `Rat`-typed `fromEvidenceQ`, and `Bucket.Timeline` discharges seven further theorems characterizing `relate`, none of them stated as a lemma in the body. 3.10's effective count and 3.11's detectability and 3.12's gap nodes carry no listing here, per 10.

The verbatim block below is `lake build`'s own output for this project, run from `lean/` with `PATH` carrying `~/.elan/bin` (Lean 4.33.1, Lake 5.0.0), captured for this pass. Lake marks each job with a Unicode check mark or warning triangle; this build has no Unicode math font either (the same gap `make_lean_snapshots.py`'s header documents), so those two glyphs are transliterated to `OK` and `WARN` below and nothing else in the line is altered:

```
OK [2/9] Built Bucket.Concept (297ms)
OK [3/9] Built Bucket.Belief (354ms)
OK [4/9] Built Bucket.Unknowns (314ms)
WARN [5/9] Built Bucket.Address (1.7s)
warning: Bucket/Address.lean:57:8: declaration uses ‘sorry‘
OK [6/9] Built Bucket.Timeline (12s)
OK [7/9] Built Bucket.Hypothesis (223ms)
OK [8/9] Built Bucket (150ms)
Build completed successfully (9 jobs).
```

The one warning is 5.2's open item: every other module builds with no warning.

B Bead list

`CROSSWALK-QUANTUM-ALGORITHM-DISCOVERY.md` §5 re-cuts every bead `HISTORY-HYPOTHESIS-ENGINE-SPEC.md`, `TIMELINE-AND- COMBINATORICS-SPEC.md`, and `IDEAL-STATE-AND-UNKNOWNS-SPEC.md` name, against what `scientific-discovery` v0.11 already solved. DROP means superseded in-spec by a later bead in the same list; KEEP means bucket-original work with no matching pattern in `scientific-discovery`; PULL means the bead builds a pattern `scientific-discovery` already solved directly inside this engine. 4 reproduces the full re-cut; the first five beads ordered for execution are `bkt-hte-evidence-span`, `bkt-hte-hypothesis-schema`, `bkt-hte-opinion-model`, `bkt-hte-concept-ontology`, and `bkt-hte-multiview-evidence`.

Table 4: Bead re-cut from `CROSSWALK-QUANTUM-ALGORITHM-DISCOVERY.md` §5. Twenty-six beads: 3 DROP, 15 KEEP, 8 PULL.

Bead	Disposition	Note
`bkt-hte-truth-score`	DROP	Superseded by `bkt-hte-opinion-model`.

Table 4 (continued)

Bead	Disposition	Note
`bkt-hte-hypothesis-schema`	KEEP	Bucket data model, no QAD analog needed.
`bkt-hte-period-model`	KEEP	History-specific period tree.
`bkt-hte-generator`	DROP	Superseded by `bkt-hte-combinatorial-generator`.
`bkt-hte-tournament`	KEEP	Elo debate ranking has no QAD counterpart.
`bkt-hte-holdout`	PULL	Adds QAD's frozen-work, sampling-strata, failures-before-repair campaign discipline to the period holdout test.
`bkt-hte-review-queue`	KEEP	Uncertainty-times-impact budgeting is bucket's own design.
`bkt-hte-ui`	KEEP	Bucket-facing route.
`bkt-hte-timeline-schema`	KEEP	Allen relations and the year axis; QAD has no dated-object algebra to pull from.
`bkt-hte-concept-ontology`	KEEP	History slot vocabulary and the five non-consensus actors.
`bkt-hte-address-scheme`	KEEP	Prime addressing is bucket's own invention; QAD uses opaque ids.
`bkt-hte-combinatorial-generator`	KEEP	Works only because history's slot vocabularies are small and closed.
`bkt-hte-kind-truth-score`	DROP	Superseded by `bkt-hte-opinion-model`.
`bkt-hte-timeline-views`	KEEP	Bucket-specific per-bin, per-event, per-pair export shape.
`bkt-hte-opinion-model`	KEEP	Subjective-logic belief has no QAD analog to pull.
`bkt-hte-stemma-dependence`	KEEP	Source-dependency stemma has no QAD analog to pull.
`bkt-hte-detectability-table`	KEEP	QAD states the absence-is-not-evidence principle but has no scaling mechanism to pull.
`bkt-hte-gap-nodes`	PULL	Merges QAD's five-factor active-retrieval priority into the gap node's VoI score.
`bkt-hte-open-world-slots`	KEEP	Dirichlet reserved mass has no QAD analog; QAD routes unknown terms to a human reviewer instead.
`bkt-hte-missing-mass`	PULL	Gates the Chao1 and Good-Turing coverage interval behind QAD's three-part audited-saturation stopping rule.
`bkt-hte-surprise-and-robustness`	KEEP	Prior-profile spread scoring has no QAD analog.
`bkt-hte-self-accounting`	PULL	Adds the target-blind generalization of QAD's quantum-blind rule to the self-report and model-prior evidence kind.
`bkt-hte-retrieval-provenance`	PULL, new	Immutable retrieval envelopes linked to a RetrievalRun, over the research gateway once mirror jobs route through it.
`bkt-hte-evidence-span`	PULL, new	Field-level evidence span on every evidence entry, in place of claim-level-only grounding.
`bkt-hte-extraction-ensemble`	PULL, new	Ensemble claim extractors plus a quality report, preserving disagreement instead of collapsing to one score.
`bkt-hte-multiview-evidence`	PULL, new	Separate similarity views feeding e_i, the blended formula kept as one view among several.

C Glossary

Every term 3 through 9 links via \term resolves to one entry below, declared once in `glossary.tex` and printed here in the order it first appears in the body.

astronomical year axis `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/TIMELINE-AND-COMBINATORICS-SPEC.md#1-timeline-data-structure`
The engine's time axis: a signed integer line with year 0 present and no gap between 1 BCE and 1 CE, unlike the historical BC/AD count. Every calendar system maps onto this axis through a converter; see 3.1.

interval with uncertainty `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/TIMELINE-AND-COMBINATORICS-SPEC.md#1-timeline-data-structure`
A dated interval on the astronomical-year axis carrying a distribution over its start and end, of type point, uniform, normal, or an OxCal posterior with its sampled density curve. 3.1.

Allen relation `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/TIMELINE-AND-COMBINATORICS-SPEC.md#1-timeline-data-structure`
One of the thirteen qualitative relations between two intervals on a time axis (before, meets, overlaps, starts, finishes, during, equals, and their inverses) [14]. The engine's only vocabulary for ordering a pair of dated hypotheses. 3.2.

concept and slot `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/TIMELINE-AND-COMBINATORICS-SPEC.md#2-concept-ontology`
A slot (ACTOR, ACTION, OBJECT, PLACE, TIME_BIN, MECHANISM, or RELATION) names one axis of a hypothesis; a concept is one member of a slot's vocabulary, carrying its own editable prior log-odds. 3.3.

prior logit `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/TIMELINE-AND-COMBINATORICS-SPEC.md#2-concept-ontology`
The log-odds a concept node carries before any hypothesis built from it sees evidence, a stored, editable number rather than a constant buried in code. 3.3.

open-world slot `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#6-unknown-unknowns`
A reserved `OTHER` concept in every slot's vocabulary, carrying a Dirichlet-process new-concept probability mass so a hypothesis can name an actor, action, or mechanism the ontology has not yet added. 7.

placement hypothesis `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/TIMELINE-AND-COMBINATORICS-SPEC.md#3-the-combinatorial-function`
A hypothesis of the form "event E happened in interval T," one point in the product space $H = \text{ACTOR} \times \text{ACTION} \times \text{OBJECT} \times \text{PLACE} \times \text{TIME\_BIN} \times \text{MECHANISM}$. 3.4.

sequence hypothesis `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/TIMELINE-AND-COMBINATORICS-SPEC.md#3-the-combinatorial-function`
A hypothesis of the form "E_1 then E_2," one point in $H_{\text{seq}} = H \times \text{ALLEN\_RELATION} \times H$: two placements joined by one of the thirteen Allen relations. 3.5.

hypothesis address `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/TIMELINE-AND-COMBINATORICS-SPEC.md#3-the-combinatorial-function`
The prime-factorization identifier of a hypothesis: each slot's vocabulary index plus one becomes the exponent of a fixed prime, and the product is the hypothesis's integer ID, decodable by trial division alone. 3.6.

evidence kind `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/TIMELINE-AND-COMBINATORICS-SPEC.md#4-truth-score-as-additive-prime-knowledge`
One of nine independent modalities evidence can carry: material or archaeological, textual or documentary, genetic, linguistic, astronomical or radiometric, geological or climate, oral tradition or myth, iconographic, or model-based inference. 3.7.

evidence tier `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/HISTORY-HYPOTHESIS-ENGINE-SPEC.md#2-truth-score-model`

A source-reliability band, T1 through T6, mapped from a source-quality score and carrying a fixed weight $k(\text{tier})$ from 2.0 down to 0.1, the maximum log-odds swing one independent piece of evidence at that tier contributes. 3.7.

evidence span `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/HISTORY-HYPOTHESIS-ENGINE-SPEC.md#3-hypothesis-data-model`
The character range and named field (`who`, `what`, `when`, `where`, `why`, or `causal_direction`) an evidence item supports, grounding a claim at the field level rather than the hypothesis as a whole. 3.7.

opinion `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#2-belief-as-opinion`
A subjective-logic tuple $\omega = (b, d, u, a)$, belief, disbelief, uncertainty mass, and base rate, with $b + d + u = 1$ [12], replacing a single posterior score. 3.8.

projected probability `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#2-belief-as-opinion`
The scalar read-out $P(h) = b(h) + a(h) \cdot u(h)$ of an opinion, collapsing belief, disbelief, and uncertainty mass into one number for ranking while the opinion itself keeps the three apart. 3.9.

base rate `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#2-belief-as-opinion`
An opinion's resting point under total ignorance, $a(h) = \text{sigmoid}(L_{\text{prior}}(h))$: what $P(h)$ equals when $b = d = 0$ and $u = 1$. 3.8.

stemma `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#3-source-dependence`
A dependency graph over evidence sources, one node per source and a directed `copies_from` or `shares_archetype_with` edge with a copy-confidence weight, borrowed from manuscript philology. 3.10.

effective count `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#3-source-dependence`
n_{eff}, the count of connected components in a stemma after pruning low-confidence edges, replacing the raw count of evidence clusters everywhere a diminishing-returns discount is applied. 3.10.

detectability `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#4-preservation-and-detectability`
$\delta(\text{period}, \text{medium}, \text{region}) \in [0, 1]$, the probability that evidence of a true hypothesis would have survived and been found in that period, medium, and region; scales absence evidence rather than treating it at a flat weight. 3.11.

gap node `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#5-known-unknowns-as-gap-nodes`
A first-class node for something the corpus has not yet looked at, an unexcavated site, an untranslated text, an undated stratum, an unread archive, naming the hypotheses it would move and its expected value of information. 3.12.

value of information `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#5-known-unknowns-as-gap-nodes`
A gap node's ranking score, folding expected uncertainty reduction, novelty, coverage gap, historical gap, and provider disagreement into one number that orders the active-learning queue. 3.12.

missing mass `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#6-unknown-unknowns`
The Good-Turing estimate f_1/N of the unseen share of the hypothesis space, where f_1 is the count of hypotheses observed in exactly one generation run and N is the total observed across runs. 3.13.

Chao1 estimator `https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#6-unknown-unknowns`
The richness estimator $S_{\text{est}} = S_{\text{obs}} + f_1^2/(2f_2)$, sizing the total hypothesis or evidence count from how many items were seen exactly once or twice across repeated runs [19, 20]. 3.13.

coverage share https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#8-coverage-claim
The fraction hypotheses_generated_and_scored(bin)/S_{est}(bin) reported with a confidence interval per time bin, replacing the claim that generation produces the full set. 19.

surprise event https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#6-unknown-unknowns
A logged event when new evidence matches no hypothesis address the generator has materialized at any slot-value combination; its rate, tracked per period, is a health signal on whether the ontology has fallen behind the evidence. 7.

prior-profile robustness https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#6-unknown-unknowns
1 minus the spread of $P(h)$ across four prior profiles, consensus, skeptic, fringe, and uniform; a hypothesis whose rank flips depending on which profile scores it earns less trust than one that does not. 7.

target-blind generation https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#7-the-model-accounting-for-itself
The rule that hypothesis generation stays complete and scored the same way regardless of which target reading, orthodox, fringe, or any consensus status between, a hypothesis would end up supporting; the history engine's own name for QAD's quantum-blind rule. 8.

ranking tournament https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/HISTORY-HYPOTHESIS-ENGINE-SPEC.md#5-hypothesis-generation-loop
The Elo-seeded, pairwise-debate ranking pass over the generated frontier, seeded from each hypothesis's posterior and recomputed against the belief model every fixed number of rounds. 8.

self-report https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/IDEAL-STATE-AND-UNKNOWNS-SPEC.md#7-the-model-accounting-for-itself
A corpus artifact every generation run emits: its assumptions, which slot vocabularies it judged incomplete, its missing-mass estimate, and its calibration score against the holdout tests. 8.

retrieval envelope https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/HISTORY-HYPOTHESIS-ENGINE-SPEC.md#8-implementation-plan
An immutable record of one fetch, linked to a `RetrievalRun` id, so a later re-fetch cannot silently change what an earlier run found without a new run id appearing in the provenance chain. 8.

neighbor mutation https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/TIMELINE-AND-COMBINATORICS-SPEC.md#3-the-combinatorial-function
The evolver's one-slot mutation move: from a hypothesis h, hold every slot fixed but one and range that slot over its vocabulary, yielding every hypothesis address one edit away from h. 5.3.

founder review https://github.com/bucket-foundation/bucket-foundation/blob/main/_intake/hypothesis-engine/HISTORY-HYPOTHESIS-ENGINE-SPEC.md#7-human-review-budget
A weekly-budgeted queue of hypotheses whose posterior falls inside a contested band or that touch a high-impact claim; a founder verdict enters as one evidence kind at the top tier, folded by fusion rather than overriding it. 9.